Problem Statement and Model Formulation

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Multi-location Newsvendor Problem

Consider a company that manages inventory in a distribution system composed of J nonidentical retailers.

- Random demand $d_j, \forall j \in [J]$, $\boldsymbol{d} = (d_j)_{j \in [J]}$.
- Contextual information $\boldsymbol{x}_j \in \mathcal{X} \subseteq \mathbb{R}^p$, such as promotions, weather and holidays. $\mathbf{X} = (\boldsymbol{x}_j)_{j \in [J]}$.
- Random variable $(\mathbf{X}, \boldsymbol{d})$ with joint distribution $\mathbb{P}$
- Conditional distribution $\mathbb{P}_{\boldsymbol{d}|\mathbf{X}}$ of $\boldsymbol{d}$ given $\mathbf{X}$.

During the selling season, demand $\boldsymbol{d}$ realizes and incurs the overage cost h_j per unit unsold inventory and lost-sale penalty cost $b_j > 0$ per unit unmet demand for retailer j. The total cost is $\sum_{j \in [J]} C_j(z_j, d_j)$, where $C_j(\boldsymbol{z}_j, \boldsymbol{d}_j)$ is:

$$C_j(z_j, d_j) = h_j(z_j - d_j)^+ + b_j(d_j - b_j)^+$$

Contextual Stochastic Problem

Suppose the planner observes covariate $\mathbf{X}$ that is predictive of demand before the decision. Then, we can solve the following contextual stochastic optimization problem to find the optimal order quantity $\boldsymbol{z}$ conditioned on observed covariate $\mathbf{X}$ (Ban and Rudin, 2019):

$$\min_{\boldsymbol{z} \geq \mathbf{0}} \mathbb{E}_{\mathbb{P}_{\boldsymbol{d}|\mathbf{X}}} \left[\sum_{j \in [J]} C_j(z_j, d_j) \mid \mathbf{X} \right] \tag{1}$$

By linearity of expectation, we can decompose the expectation to:

$$\min_{\boldsymbol{z} \geq \mathbf{0}} \sum_{j \in [J]} \mathbb{E}_{\mathbb{P}^{(j)}_{\boldsymbol{d}|\mathbf{X}}} \left[C_j(z_j, d_j) \mid \mathbf{X} \right]$$

where $\mathbb{P}^{(j)}_{\boldsymbol{d}|\mathbf{X}}$ denotes the marginal distribution of d_j in $\mathbb{P}_{\boldsymbol{d}|\mathbf{X}}$. However, we do not have access to true distribution.

Separate Data Sets (Multi-source Predit-then-Optimize)

Instead of assuming an aggregated data set at disposal (Notz and Pibernik, 2022), we consider the setting that the decision maker has access to J disjoint data sets collected by retailers individually.

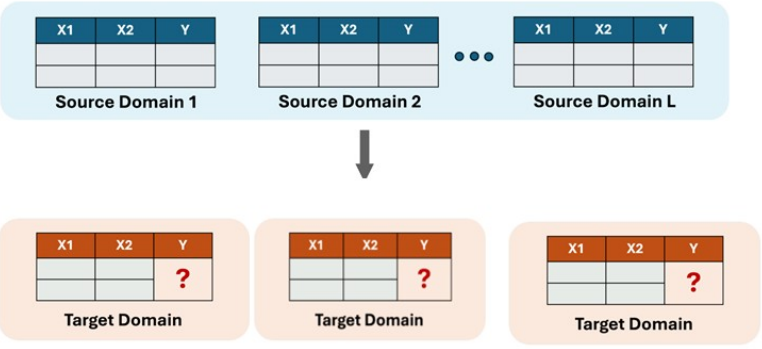
- Each retailer j is associated with a **separate data set** $S^{(j)} = \{(\boldsymbol{x}_i^{(j)}, d_i^{(j)})\}_{i=1}^{N_j}$, where $(\boldsymbol{x}_i^{(j)}, d_i^{(j)})$ is i.i.d. samples generated from a distinct distribution:

$$\boldsymbol{x}_i^{(j)} \sim \mathbb{P}_{\mathbf{X}}^{(j)}, d_i^{(j)} \mid \boldsymbol{x}_i^{(j)} \sim \mathbb{P}_{\boldsymbol{d}|\mathbf{X}}^{(j)}, \quad \forall j \in [J]$$

  where $\mathbb{P}_{\mathbf{X}}^{(j)}$ denotes the distribution of $\boldsymbol{x}_j$, and $\mathbb{P}_{\boldsymbol{d}|\mathbf{X}}^{(j)}$ denotes the marginal distribution of $\boldsymbol{d} \mid \mathbf{X}$ for retailer j.
- $\mathbf{D}^{(j)} = (d_1^{(j)}, \ldots, d_{N_j}^{(j)}) \in \mathbb{R}^{N_j}$, $\mathbf{X}^{(j)} = (\boldsymbol{x}_1^{(j)}, \ldots, \boldsymbol{x}_{N_j}^{(j)})^\top \in \mathbb{R}^{N_j \times p}$ and total sample size $N = \sum_{j \in [J]} N_j$.
- **Heterogenous Distibution**: $\mathbb{P}_{\mathbf{X}}^{(j)}$ and $\mathbb{P}_{\boldsymbol{d}|\mathbf{X}}^{(j)}$ can be heterogenous among retailers.
- **Local data set** $\cup_{j \in [J]} S^{(j)}$, **Global data set** $\cup_{j \in [J]} S^{(j)}$.

Separate Data Sets (Multi-source Predit-then-Optimize)

- **Our Framework:**



Each target domain is a mixture of other distributions, the weight determined by learning; then construct DRO set.

Separate Data Sets (Multi-source Predit-then-Optimize)

The decision is made in two steps: First, we estimate the conditional distribution $\mathbb{P}_{\boldsymbol{d}|\mathbf{X}}$ given covariate $\mathbf{X}$. Second, we adopt a DRO approach to find a decision immunized against distribution ambiguity.

To obtain an estimate of marginal distribution $\mathbb{P}^{(j)}_{\boldsymbol{d}|\mathbf{X}}$, we assume the demand d and the covariates $\boldsymbol{x}$ of a single retailer is governed by:

- **Global model**: shared across all retailers

$$d_{\text{global}} = \boldsymbol{x}^\top \boldsymbol{\beta}_{\text{global}} + \varepsilon_{\text{global}}$$

- **Local model**: specific to each retailer j

$$d^{(j)}_{\text{local}} = \boldsymbol{x}^\top \boldsymbol{\beta}^{(j)}_{\text{local}} + \varepsilon^{(j)}_{\text{local}}$$

- **Assumption**: $x^1 = 1$ to allow for an intercept term, $\boldsymbol{\beta}_{\text{global}}, \boldsymbol{\beta}^{(j)}_{\text{local}} \in \mathbb{R}^p$ are the nonzero linear regression coefficients with $\|\boldsymbol{\beta}_{\text{global}}\|_2 \leq B$ and $\|\boldsymbol{\beta}^{(j)}_{\text{local}}\|_2 \leq B$, and $\varepsilon_{\text{global}}, \varepsilon^{(j)}_{\text{local}}$ are bounded random noise with zero mean, independent of covariate $\boldsymbol{x}$. Note that we do not impose that $\varepsilon^{(j)}_{\text{local}}$ is independent over $j \in [J]$.

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Linear Demand Model

2 Joint Estimator

The global model and local models are **closely related**, in the sense that they follow similar demand pattern in a market.

$$\boldsymbol{\beta}_{\text{local}}^{(j)} = \boldsymbol{\beta}_{\text{global}} + \xi^{(j)}$$

where $\xi^{(j)}$ is the **bias** between the global model and local model. This relation indicates that we can leverage global data set $\cup_{j\in[J]} S^{(j)}$ to help predict the local demand $d_{\text{local}}^{(j)}$ with careful debias.

If the true parameters $\boldsymbol{\beta}_{\text{local}}^{(j)}$ and $\varepsilon_{\text{local}}^{(j)}$ are known, then the conditional stochastic program (1) is equivalent to:

$$\boldsymbol{v}^*(\mathbf{X}) = \min_{z_j \geq 0, j\in[J]} \sum_{j=1}^{J} \mathbb{E}\left[C_j\left(\boldsymbol{z}_j, \boldsymbol{x}_j^\top \boldsymbol{\beta}_{\text{local}}^{(j)} + \varepsilon_{\text{local}}^{(j)}\right)\right] \tag{2}$$

where the expectation is taken with respect to the distribution of $\varepsilon_{\text{local}}^{(j)}$. We cannot solve (2) directly because the $\boldsymbol{\beta}_{\text{local}}^{(j)}$ and $\varepsilon_{\text{local}}^{(j)}$ are unknown.

This can be extended to any machine learning method (random forest, Kernel density estimation).

Empirical Optimization Problem / Contextual Stochastic Optimization

2 Joint Estimator

- **Empirical Residual**: For a given $\boldsymbol{\beta}$, we define the following mapping $\hat{\varepsilon}(\boldsymbol{\beta}, j, i)$:

$$\hat{\varepsilon}(\boldsymbol{\beta}, j, i) = d_i^{(j)} - \mathbf{x}_i^{(j)\top}\boldsymbol{\beta}$$

  which show that empirical residual is dependent on the regression coefficient $\boldsymbol{\beta}$.
- **Predicted Demand Sample**: we also define the following mapping $\hat{d}$ to predict the demand at any retailer:

$$\hat{d}(\boldsymbol{\beta}, \mathbf{x}, j, i) = \mathbf{x}^\top\boldsymbol{\beta} + \hat{\varepsilon}(\boldsymbol{\beta}, j, i) = (\mathbf{x} - \mathbf{x}_i^{(j)})^\top\boldsymbol{\beta} + d_i^{(j)}$$

  The predicted demand is obtained by adjusting the historical demand at $\mathbf{x}_i^{(j)}$ with the predicted demand change induced by covariate shift $(\mathbf{x} - \mathbf{x}_i^{(j)})$.
- **Predicted Empirical Distribution**: for given $\boldsymbol{\beta}$ and covariate $\mathbf{x}$, we can construct the predicted empirical distribution of local demand d_j, using empirical residuals on the data set $S^{(j)}$:

$$\hat{\mathbb{P}}(\boldsymbol{\beta}, \mathbf{x}, j) = \frac{1}{N_j}\sum_{i=1}^{N_j}\delta_{\hat{d}(\boldsymbol{\beta}, \mathbf{x}, j, i)}$$

  where $\delta_{\hat{d}(\boldsymbol{\beta}, \mathbf{x}, j, i)}$ is a Dirac measture that places a unit mass at $\hat{d}(\boldsymbol{\beta}, \mathbf{x}, j, i)$.

Identification Assumption

Our goal is to estimate $\beta^{(j)}_{\text{local}}$ and $\varepsilon^{(j)}_{\text{local}}$ accurately from separate data sets $S^{(j)}$ for $j \in [J]$. Since we have a linear model, the **ordinary least square (OLS) estimator** is the most prevalent choice Bastani (2021). We make some assumptions to assure idenfication of regression coefficients:

- **Standardization:** We standardize each column $\boldsymbol{x}_i^{(j)}$ of $\mathbf{X}^{(j)}$ such that $\|\boldsymbol{x}_i^{(j)}\|_2^2 = N_j$. We further define the *sample covariance matrix* for each j as
$$\Sigma^{(j)} = \frac{1}{N_j}\mathbf{X}^{(j)\top}\mathbf{X}^{(j)}.$$
The diagonal of $\Sigma^{(j)}$ satisfies $\mathrm{Diag}(\Sigma^{(j)}) = \mathbf{1} \in \mathbb{R}^p$.
This is a standard practice to normalize covariates when using regularized regression.
- **Positive definiteness:** The sample covariance matrix $\Sigma^{(j)} = \frac{1}{N_j}\mathbf{X}^{(j)\top}\mathbf{X}^{(j)}$ is positive definite, with the smallest eigenvalue bounded below by $\psi > 0$.
The second assumption is standard in statistics for idenfication for local and global estimator.
- **Sparsity:** The bias vector $\xi^{(j)}$ is sparse; that is, $\|\xi^{(j)}\|_0 = s \ll p$ for each j.
The joint estimator of Bastani (2021) is most effective when sparsity is present.

Global and Local Estimator

Our goal is to estimate $\beta^{(j)}_{\text{local}}$ and $\varepsilon^{(j)}_{\text{local}}$ accurately from separate data sets $S^{(j)}$ for $j \in [J]$. There are two natural way to do so:

- **Local Regression**. To solve a regression problem for each local data set $S^{(j)}$.

$$\hat{\beta}_{\text{local}} = \arg\min_{\beta} \left\{ \frac{1}{N_j} \|\mathbf{D}^{(j)} - \mathbf{X}^{(j)\top}\beta\|_2^2 \right\}$$

  Drawback: local data sets often face data scarcity, which may lead to high-variance estimation.

- **Global Regression**. To obtain a better generalization performance, an alternative way is to collaboratively train a single global estimator model on the global data set $\cup_{j\in[J]} S^{(j)}$.

$$\hat{\beta}_{\text{global}} = \arg\min_{\beta} \left\{ \sum_{j\in[J]} \frac{1}{N_j} \|\mathbf{D}^{(j)} - \mathbf{X}^{(j)\top}\beta\|_2^2 \right\}$$

  Drawback: However, such a global model may perform well on average but generalize poorly for a particular retailer j, because there are distributional heterogeneity (Guo et al., 2025; Mo et al., 2024).

Global and Local Estimator

2 Joint Estimator

We can define the residual distribution by global estimator and local estimator

- Global Residual distribution for retailer j:

$$\hat{\mathbb{P}}(\hat{\boldsymbol{\beta}}_{\text{global}}, \boldsymbol{x}, j) = \frac{1}{N_j} \sum_{i=1}^{N_j} \delta_{\hat{d}(\hat{\boldsymbol{\beta}}_{\text{global}}, \boldsymbol{x}, j, i)}$$

- Local Residual distribution:

$$\hat{\mathbb{P}}(\hat{\boldsymbol{\beta}}_{\text{local}}, \boldsymbol{x}, j) = \frac{1}{N_j} \sum_{i=1}^{N_j} \delta_{\hat{d}(\hat{\boldsymbol{\beta}}_{\text{local}}, \boldsymbol{x}, j, i)}$$

where $\delta_{\hat{d}(\boldsymbol{\beta}, \boldsymbol{x}, j, i)}$ is a Dirac measture that places a unit mass at $\hat{d}(\boldsymbol{\beta}, \boldsymbol{x}, j, i)$.

Joint Estimator (Regularized Regression)

We adapt the joint estimator from Bastani (2021) on the global and local data as follows:

- **Step 1: Global Estimator** We run an OLS regression on the global data set $\bigcup_{j\in[J]} S^{(j)}$ and obtain a *global estimator* $\hat{\beta}_{\text{global}}$ defined as

$$\hat{\beta}_{\text{global}} = \arg\min_{\beta} \left\{ \sum_{j\in[J]} \frac{1}{N_j} \left\| \mathbf{D}^{(j)} - \mathbf{X}^{(j)\top}\beta \right\|_2^2 \right\}. \tag{3}$$

- **Step 2: Joint Estimator** For each data set $S^{(j)}$, we obtain a joint estimator $\hat{\beta}^{(j)}_{\text{joint}}$ via a LASSO regression. The regularization term is based on the deviation between the estimated coefficient $\hat{\beta}^{(j)}_{\text{joint}}$ and the global estimator $\hat{\beta}_{\text{global}}$. The joint estimator is a **LASSO estimator of Local Data debiased by Global Estimator**.

$$\hat{\beta}^{(j)}_{\text{joint}} = \arg\min_{\beta} \left\{ \frac{1}{N_j} \left\| \mathbf{D}^{(j)} - \mathbf{X}^{(j)\top}\beta \right\|_2^2 + \lambda^{(j)} \left\| \beta - \hat{\beta}_{\text{global}} \right\|_1 \right\}. \tag{4}$$

  Here, $\lambda^{(j)} > 0$ is a **regularization parameter** that controls the influence of the global estimator $\hat{\beta}_{\text{global}}$ in the joint model. The **ℓ_1-norm penalty exploits the sparsity structure** of the bias term $\xi^{(j)}$.

Empirical Optimization Problem

2 Joint Estimator

If all the joint estimators $\hat{\boldsymbol{\beta}}^{(j)}_{\text{joint}}$ are known, the empirical optimization problem corresponding to Problem (2) is defined as

$$\begin{aligned}\hat{\boldsymbol{v}}^{\text{ER}}(\mathbf{X}) &= \min_{\boldsymbol{z}\geq \mathbf{0}} \sum_{j\in[J]} \mathbb{E}\left[C_j(z_j, d_j) \mid \mathbf{X}\right] \\ &= \min_{z_j\geq 0, j\in[J]} \sum_{j=1}^{J} \frac{1}{N_j} \sum_{i=1}^{N_j} C_j\left(\boldsymbol{z}_j, \hat{d}(\hat{\boldsymbol{\beta}}^{(j)}_{\text{joint}}, \boldsymbol{x}_j, j, i)\right)\end{aligned} \tag{5}$$

where the expectation is taken with respect to the predicted empirical distribution $\hat{\mathbb{P}}(\hat{\boldsymbol{\beta}}^{(j)}_{\text{joint}}, \boldsymbol{x}_j, j)$ (Kannan et al., 2025).

Limitation: A limitation of the empirical optimization approach may exhibit disappointing out-of-sample performance when the sample size is limited (Kannan et al., 2024). To address such small-data issue, we develop a DRO model to leverage predicted empirical distributions from multiple data sets, ensuring generalization to the specific retailer j.

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1. Variation Regularization for cost function

3 Method 1: Regularization for Contextual Stochastic Optimization

Inspired by Joint Estimator in learning stage (Bastani, 2021), we can extend this regularization technique into Contextual Stochastic Optimization. Gao et al. (2024) proposed a Wasserstein DRO and Variation Regularization:

$$\min_{f \in \mathcal{F}} \{\mathbb{E}_{z \sim \mathbb{P}_n}[f(z)] + \rho_n \mathcal{V}_{\mathbb{P}_n,q}(f)\}$$

where ρ_n is regularization parameter, and $\mathcal{V}_{\mathbb{P}_n,q}(f)$ is defined as the variation of cost function f with respect to an empirical distribution $\mathbb{P}_n$. Intuitively, it is **the variation magnitude of f under the distribution $\mathbb{P}_n$**.

When the variation of loss $\mathcal{V}_{\mathbb{P}_n,q}(f)$ is controlled, small perturbations of random data would have little impact on the expected loss and thus would not deteriorate the solution quality much. So We can bound **the difference of variation of loss** on global distribution $\mathcal{V}_{\mathbb{P}_{global},q}(f)$ and local distribution $\mathcal{V}_{\mathbb{P}_n,q}(f)$.

$$\min_{f \in \mathcal{F}} \mathbb{E}_{z \sim \mathbb{P}_n}[f(z)] + \rho |\mathcal{V}_{\mathbb{P}_{global}}(f) - \mathcal{V}_{\mathbb{P}_n}(f)|$$

Sinkhorn DRO adopts the Sinkhorn distance to construct the ambiguity set (Wang et al., 2025a). The Sinkhorn distance can be viewed as a **smoothed version** of the Wasserstein distance. It is defined as the minimum transport cost between two distributions with **entropic regularization**.

DEFINITION 1 (SINKHORN DISTANCE). Let $\mathcal{Z}$ be a measurable set. Consider distributions $\mathbb{P}, \mathbb{Q} \in \mathcal{P}(\mathcal{Z})$, and let $\mu, \nu \in \mathcal{M}(\mathcal{Z})$ be two reference measures such that $\mathbb{P} \ll \mu$, $\mathbb{Q} \ll \nu$. For regularization parameter $\epsilon \geq 0$, the *Sinkhorn distance* between two distributions $\mathbb{P}$ and $\mathbb{Q}$ is defined as

$$\mathcal{W}_\epsilon(\mathbb{P}, \mathbb{Q}) = \inf_{\gamma \in \Gamma(\mathbb{P}, \mathbb{Q})} \left\{ \mathbb{E}_{(x,y)\sim\gamma}[c(x,y)] + \epsilon H(\gamma \mid \mu \otimes \nu) \right\},$$

where $\Gamma(\mathbb{P}, \mathbb{Q})$ denotes the set of joint distributions whose first and second marginal distributions are $\mathbb{P}$ and $\mathbb{Q}$ respectively, $c(x,y)$ denotes the transport cost, and $H(\gamma \mid \mu \otimes \nu)$ denotes the relative entropy of γ with respect to the product measure $\mu \otimes \nu$:

$$H(\gamma \mid \mu \otimes \nu) = \mathbb{E}_{(x,y)\sim\gamma} \left[\log \left(\frac{\mathrm{d}\gamma(x,y)}{\mathrm{d}\mu(x)\,\mathrm{d}\nu(y)} \right) \right],$$

where $\frac{\mathrm{d}\gamma(x,y)}{\mathrm{d}\mu(x)\,\mathrm{d}\nu(y)}$ stands for the density ratio of γ with respect to $\mu \otimes \nu$ evaluated at (x,y). $\diamondsuit$

Regularization Term: penalizes highly concentrated couplings, leading to more **smooth, stable** joint distribution γ.

$$\mathcal{W}_\epsilon^{\text{Ent}}(\mathbb{P}, \mathbb{Q}) = \mathcal{W}_\epsilon(\mathbb{P}, \mathbb{Q}) + \mathbb{E}_{(x,y)\sim\gamma}\left[\log\left(\frac{\mathrm{d}\gamma(x,y)}{\mathrm{d}\mu(x)\mathrm{d}\nu(y)}\right)\right] \tag{6}$$

Using global estimation as a regularization reference: We can extend this idea, to choose reference measures μ as residual distribution from global estimator, i.e., $\hat{\mathbb{P}}_{\text{global}}$:

$$H(\gamma \mid \hat{\mathbb{P}}_{\text{global}} \otimes \nu) = \mathbb{E}_{(x,y)\sim\gamma}\left[\log\left(\frac{\mathrm{d}\gamma(x,y)}{\mathrm{d}\hat{\mathbb{P}}_{\text{global}}\mathrm{d}\nu(y)}\right)\right],$$

where $\hat{\mathbb{P}}_{\text{global}} \ll \mu$.

This should entropy term penalizes couplings that deviate too far from the product baseline $P_{\text{global}} \otimes \nu$, Thus, the global estimator acts as a debias term in distributional distance.

Disadvantage: The distributions must be absolutely continuous to reference measures.

Then we can define a Sinkhorn DRO model as:

$$\min_{\boldsymbol{z}\geq 0} \sup_{\mathbb{P}_{\boldsymbol{d}|\mathbf{X}}\in\mathbb{B}_{\rho,\epsilon}(\widehat{\mathbb{P}},\hat{\mathbb{P}}_{\text{global}})} \mathbb{E}_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}}\left[\sum_{j\in[J]} C_j(z_j, d_j) \mid \mathbf{X}\right] \tag{Sinkhorn}$$

where $\mathbb{B}_{\rho,\epsilon}(\widehat{\mathbb{P}}, \hat{\mathbb{P}}_{\text{global}})$ is a Sinkhorn-ball of radius ρ centered at the nominal distribution $\hat{\mathbb{P}}$ **regularized by deviation from** $\hat{\mathbb{P}}_{\text{global}}$.

$$\mathbb{B}_{\rho,\epsilon}(\widehat{\mathbb{P}}, \mathbb{P}) := \left\{\mathbb{P} : \mathcal{W}_\epsilon(\widehat{\mathbb{P}}, \mathbb{P}) \leq \rho\right\}$$

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Mixture DRO Approach

We borrow ideas from the conditional group DRO framework and federate learning (Marfoq et al., 2021; Wang et al., 2025b), to define **the local distribution as mixtures from the predicted empirical distributions**.

Based on input covariate $\mathbf{X}$, we define the mixture conditional distribution $\mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}}$ as:

$$\mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}} = \sum_{l \in [J]} q_l^{(j)} \hat{\mathbb{P}}(\hat{\boldsymbol{\beta}}^{(l)}_{\text{joint}}, \boldsymbol{x}_j, l)$$

where $\boldsymbol{q}^{(j)} \in \Delta^J$ and Δ^J denotes $(J-1)$-dimensional probability simplex. We first **fix the input covariate $\mathbf{X}$**, and estimate the empirical distribution $\hat{\mathbb{P}}(\hat{\boldsymbol{\beta}}^{(l)}_{\text{joint}}, \mathbf{x}_j, l)$ **using different data set S^l for $l \in [J]$**.

Mixture DRO Approach

We borrow ideas from the conditional group DRO framework and feaderate learning (Marfoq et al., 2021; Wang et al., 2025b), to define **the local distribution as mixtures from the predicted empirical distributions**.

We can construct an ambiguity set d_j conditioning on the observed covariate $\mathbf{X}$:

$$\mathcal{F}_j(\mathbf{X}, \theta) = \left\{ \mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}} \in \mathcal{P}(\mathbb{R}_+) \;\middle|\; \begin{array}{l} \boldsymbol{d}_j \sim \mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}} \\ \mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}} = \sum_{l \in [J]} q^{(j)}_l \hat{\mathbb{P}}(\hat{\boldsymbol{\beta}}^{(l)}_{\text{joint}}, \boldsymbol{x}_j, \boldsymbol{l}) \\ \boldsymbol{q}^{(j)} \in \mathcal{H}(\hat{\boldsymbol{q}}^{(j)}, \theta) \end{array} \right\}$$

where the uncertainty set $\mathcal{H}(\hat{\boldsymbol{q}}, \theta) \subseteq \Delta^J$ is defined as:

$$\mathcal{H}(\hat{\boldsymbol{q}}, \theta) = \left\{ \boldsymbol{q} \in \mathbb{R}^J \;\middle|\; \begin{array}{l} \mathbf{1}^\top \boldsymbol{q} = 1 \\ \boldsymbol{q} \geq \mathbf{0} \\ \|\boldsymbol{q} - \hat{\boldsymbol{q}}\| \leq \theta \end{array} \right\}$$

- **Mixture weight: $\boldsymbol{q}$ reside in the θ-neighborhood of $\hat{\boldsymbol{q}}$** with respect to the general norm.
- **Prior knowledge: $\hat{\boldsymbol{q}}$ is a pre-specified weight** obtained from training.
- **Robustness budget:** $\theta \geq 0$ controls the size of $\mathcal{H}(\hat{\boldsymbol{q}}, \theta)$.

Mixture DRO Approach

Then, the ambiguity set for the joint distribution of demand $\boldsymbol{d}$ given covariate $\mathbf{X}$ is:

$$\mathcal{F}(\mathbf{X}, \theta) = \left\{ \mathbb{Q}_{\boldsymbol{d}|\mathbf{X}} \in \mathcal{P}_0(\mathbb{R}^J_+) \;\middle|\; \begin{array}{ll} (d_1, \ldots, d_J) \sim \mathbb{Q}_{\boldsymbol{d}|\mathbf{X}} & \\ d_j \sim \mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}} & \forall j \in [J] \\ \mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}} \in \mathcal{F}_j(\mathbf{X}, \theta) & \forall j \in [J] \end{array} \right\} \tag{Mixture}$$

With the defined ambiguity set and observed covariate $\mathbf{X}$, we formulate the following contextual DRO model to minimize the worst-case expected total cost:

$$\hat{\boldsymbol{v}}^{\text{DRO}}(\mathbf{X}) = \min_{\boldsymbol{z} \geq \mathbf{0}} \sup_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}} \in \mathcal{F}(\mathbf{X}, \theta)} \mathbb{E}_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}} \left[\sum_{j \in [J]} C_j(z_j, d_j) \mid \mathbf{X} \right] \tag{DRO}$$

We also denote an optimal solution to Problem (DRO) as $\hat{\boldsymbol{z}}^{\text{DRO}}(\mathbf{X})$.

Problem Reformulation

4 Method 2: Mixture DRO Approach

We first consider the classic DRO formulation.

$$\hat{\boldsymbol{v}}^{\text{DRO}}(\mathbf{X}) = \min_{\boldsymbol{z}\geq \mathbf{0}} \sup_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}\in\mathcal{F}(\mathbf{X},\theta)} \mathbb{E}_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}}\left[\sum_{j\in[J]} C_j(z_j, d_j) \mid \mathbf{X}\right] \tag{DRO}$$

Given $\mathcal{F}(\mathbf{X},\theta)$ in (Mixture), we have equivalently(from Proposition 1 in Hu et al. (2025)):

$$\hat{\boldsymbol{v}}^{\text{DRO}}(\mathbf{X}) = \begin{cases} \min\limits_{\boldsymbol{\tau},\boldsymbol{z}} & \sum\limits_{j\in[J]} \tau_j \\ \text{s.t.} & \sup\limits_{\mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}}\in\mathcal{F}_j(\mathbf{X},\theta)} \mathbb{E}_{\mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}}}\left[C_j(z_j, d_j) \mid \mathbf{X}\right] \leq \tau_j, \quad \forall j\in[J], \\ & \boldsymbol{z}\geq \mathbf{0}, \quad \boldsymbol{\tau}\geq \mathbf{0}. \end{cases} \tag{7}$$

Problem Reformulation

4 Method 2: Mixture DRO Approach

- By linearity of expectation, we can write (DRO) as:

$$\hat{\boldsymbol{v}}^{\text{DRO}}(\mathbf{X}) = \min_{\boldsymbol{z}\geq \mathbf{0}} \sum_{j\in[J]} \sup_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}\in\mathcal{F}(\mathbf{X},\theta)} \mathbb{E}_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}}\left[C_j(z_j, d_j) \mid \mathbf{X}\right]$$

- We introduce epigraphical variable τ_j for all $j \in [J]$ to represent the worst-case expected cost for retailer j.
- By definition of (Mixture) ambiguity set, for any $\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}} \in \mathcal{F}(\mathbf{X}, \theta)$, there exist $\mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}} \in \mathcal{F}_j(\mathbf{X}, \theta), \forall j \in [J]$ to constitute the joint distribution $\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}$.

$$\sup_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}\in\mathcal{F}(\mathbf{X},\theta)} \mathbb{E}_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}}\left[C_j(z_j, d_j) \mid \mathbf{X}\right] \leq \sup_{\mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}}\in\mathcal{F}_j(\mathbf{X},\theta)} \mathbb{E}_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}}\left[C_j(z_j, d_j) \mid \mathbf{X}\right]$$

- Conversely, for any $\mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}} \in \mathcal{F}_j(\mathbf{X}, \theta), \forall j \in [J]$, we can construct a joint distribution $\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}} \in \mathcal{F}(\mathbf{X}, \theta)$ so that:

$$\sup_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}\in\mathcal{F}(\mathbf{X},\theta)} \mathbb{E}_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}}\left[C_j(z_j, d_j) \mid \mathbf{X}\right] \leq \sup_{\mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}}\in\mathcal{F}_j(\mathbf{X},\theta)} \mathbb{E}_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}}\left[C_j(z_j, d_j) \mid \mathbf{X}\right]$$

This completes the proof.

Problem Reformulation

4 Method 2: Mixture DRO Approach

We next aim to transform the Problem (6) into a convex optimization problem. We consider the reformulation of the following constraint:

$$\sup_{\mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}} \in \mathcal{F}_j(\mathbf{X},\theta)} \mathbb{E}_{\mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}}} \left[C_j(z_j, d_j) \mid \mathbf{X} \right] \leq \tau_j \tag{8}$$

For notational simplicity, we drop the subscript j and write $\hat{\mathbb{P}}^{(j)}_l = \hat{\mathbb{P}}(\hat{\boldsymbol{\beta}}^{(l)}_{\text{joint}}, \mathbf{x}_j, l)$. By definition, **$\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}$ is a mixture distribution of $\hat{\mathbb{P}}(\hat{\boldsymbol{\beta}}^{(l)}_{\text{joint}}, \mathbf{x}, l)$**.

$$\mathbb{E}_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}} \left[C(\boldsymbol{z}, d) \mid \mathbf{X} \right] = \sum_{l \in [J]} q_l \mathbb{E}_{\hat{\mathbb{P}}_l} [C(\boldsymbol{z}, d)]$$

Then the left-hand term can be simplified as

$$\begin{cases} \sup\limits_{\boldsymbol{q}} & \sum\limits_{l \in [J]} q_l \, \mathbb{E}_{\hat{\mathbb{P}}_l} [C(\boldsymbol{z}, d)] \\ \text{s.t.} & \mathbf{1}^\top \boldsymbol{q} = 1, \\ & \|\boldsymbol{q} - \hat{\boldsymbol{q}}\| \leq \theta, \\ & \boldsymbol{q} \geq \mathbf{0}. \end{cases}$$

Problem Reformulation

4 Method 2: Mixture DRO Approach

By the definition of dual norm, we substitute the norm constraint by the following:

$$\boldsymbol{\gamma}^\top(\boldsymbol{q}-\hat{\boldsymbol{q}}) \le \theta, \quad \forall \boldsymbol{\gamma} \text{ such that } \|\boldsymbol{\gamma}\|_* \le 1.$$

Then we can write the Lagrangian of the problem by introducing dual variable $\lambda \in \mathbb{R}$ and $\boldsymbol{\nu} \in \mathbb{R}^J_+$:

$$L(\boldsymbol{q}, \lambda, \boldsymbol{\nu}, \boldsymbol{\gamma}) = \mathbf{E}^\top \boldsymbol{q} + \lambda(1 - \mathbf{1}^\top \boldsymbol{q}) + \theta - \boldsymbol{\gamma}^\top(\boldsymbol{q}-\hat{\boldsymbol{q}}) + \boldsymbol{\nu}^\top \boldsymbol{q}$$

where **we use $\mathbf{E}^\top \boldsymbol{q}$ to denote** $\sum_{l\in[J]} q_l \mathbb{E}_{\hat{\mathbb{P}}_l}[\mathcal{C}(\boldsymbol{z}, d)]$ for short. By rearranging the terms, we have:

$$L(\boldsymbol{q}, \lambda, \boldsymbol{\nu}, \boldsymbol{\gamma}) = (\mathbf{E} - \lambda\mathbf{1} - \boldsymbol{\gamma} + \boldsymbol{\nu})^\top \boldsymbol{q} + \lambda + \theta - \boldsymbol{\gamma}^\top \hat{\boldsymbol{q}}$$

The result dual problem is:

$$\begin{cases} \inf\limits_{\lambda, \boldsymbol{\nu} \ge \mathbf{0}} & \lambda + \theta - (\mathbf{E} - \lambda\mathbf{1} + \boldsymbol{\nu})^\top \hat{\boldsymbol{q}} \\ \text{s.t.} & \|\mathbf{E} - \lambda\mathbf{1} + \boldsymbol{\nu}\|_* \le 1. \end{cases}$$

Problem Reformulation

4 Method 2: Mixture DRO Approach

Given $\mathcal{F}(\mathbf{X}, \theta)$ in (Mixture), the Problem (DRO) is equivalent to the following convex optimization problem:

$$\hat{\boldsymbol{v}}^{\mathrm{DRO}}(\mathbf{X}) = \begin{cases} \min\limits_{\boldsymbol{\tau}, \mathbf{z}, \{\lambda_j, \boldsymbol{\nu}_j\}_{j=1}^{J}} & \sum\limits_{j \in [J]} \tau_j \\ \text{s.t.} & \lambda_j - (\mathbf{E}_j - \lambda_j \mathbf{1} + \boldsymbol{\nu}_j)^\top \hat{\boldsymbol{q}} + \theta \leq \tau_j, \quad \forall j \in [J], \\ & \|\mathbf{E}_j - \lambda_j \mathbf{1} + \boldsymbol{\nu}_j\|_* \leq 1, \quad \forall j \in [J], \\ & \lambda_j \in \mathbb{R}, \quad \boldsymbol{\nu}_j \geq \mathbf{0}, \quad \forall j \in [J], \\ & \mathbf{z} \geq \mathbf{0}, \quad \boldsymbol{\tau} \geq \mathbf{0}. \end{cases} \tag{9}$$

- **Advantage of Empirical distribution**: Note that we write $\mathbf{E}_j = (\mathbb{E}_{\hat{\mathbb{P}}_l^{(j)}}[C_j(z_j, d_j)])_{l \in [J]}$. The expectation over empirical distribution is computationally tractable:

$$\mathbb{E}_{\hat{\mathbb{P}}_l^{(j)}}[C_j(z_j, d_j)] = \frac{1}{N_j}[C_j(z_j, d_j)] = \frac{1}{N_j} \sum_{i=1}^{N_j} C_j\left(\mathbf{z}_j, \hat{d}(\hat{\boldsymbol{\beta}}_{\text{joint}}^{(j)}, \mathbf{x}_j, j, i)\right)$$

- **Disadvantage of Empirical distribution**: the empirical distribution is finitely supported, so the worst-case distribution is a discrete distribution. When the underlying true distribution is continuous, this may induce overly conservative solutions.

Potential Outcomes

4 Method 2: Mixture DRO Approach

- **Conjecture: z^* for Problem (DRO) for any retailer is a weighted average of individual empirical newsvendor solution**. There may exist **closed-form solution** that for some $j \in [J]$, $\boldsymbol{q}^* \in \mathcal{H}(\hat{\boldsymbol{q}}, \theta)$:

$$z^* = \arg\min_z \sum_{l \in [J]} q_l^* \mathbb{E}_{\hat{\mathbb{P}}_l}[\mathcal{C}(\boldsymbol{z}, d)] = \sum_{l \in [J]} q_l^* z^{(l)}$$

where $z^{(l)}$ **is empirical newsvendor solution for $l-$th SAA**:

$$z^{(l)} = \arg\min_z \mathbb{E}_{\hat{\mathbb{P}}_l}[\mathcal{C}(\boldsymbol{z}, d)] = \inf\left\{z : \hat{F}_l(z) \geq \frac{b}{b+h}\right\}$$

$\hat{F}_l$ is the empirical cumulative distribution function (cdf) of the demand from the $l-th$ dataset.

Mixture DRO Approach

- **Extension 1: Budget-driven DRO** (Hu et al., 2025). Given a budget τ_j for each j and maximize the robustness budget θ.

$$\theta^*(\boldsymbol{\tau}) = \begin{cases} \sup\limits_{\theta} & \theta \\ \text{s.t.} & \sup\limits_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}} \in \mathcal{F}(\mathbf{X},\theta)} \mathbb{E}_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}}\left[C_j(\boldsymbol{z}_j, d_j) \mid \mathbf{X}\right] \leq \tau_j, \quad \forall j \in [J], \\ & \theta \geq 0. \end{cases}$$

- **Extension 2: Robust Satisficing** (Sim et al., 2024). Given a budget τ_j for each j, and minimize the fragility measure. We can further account for the uncertainty in regression coefficients $\boldsymbol{\beta}$.

$$\kappa^*(\boldsymbol{\tau}) = \begin{cases} \min\limits_{\kappa} & \kappa \\ \text{s.t.} & \mathbb{E}_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}}\left[C_j(\boldsymbol{z}_j, d_j) \mid \mathbf{X}\right] - \tau_j \leq \kappa(\|\boldsymbol{q} - \hat{\boldsymbol{q}}\| - \theta)^+, \quad \forall \mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}, j \in [J], \\ & \kappa \geq 0. \end{cases}$$

- **Extension 3: Minimum volume confidence set** (Lou et al., 2024): $\mathcal{H}(\hat{\boldsymbol{q}}, \theta) \Rightarrow \mathcal{H}(\hat{\boldsymbol{q}}, \mathbf{D}, \theta)$.

$$\mathcal{H}(\hat{\boldsymbol{q}}, \mathbf{D}, \theta) = \mathcal{H}(\hat{\boldsymbol{q}}, \theta) = \left\{ \boldsymbol{q} \in \Delta^L \,\middle|\, \|\mathbf{D}^-(\boldsymbol{q} - \hat{\boldsymbol{q}})\| \leq \theta \right\}$$

- **Extension 3: Minimum volume confidence set (MVCS)** (Lou et al., 2024): $\mathcal{H}(\hat{\boldsymbol{q}}, \theta) \Rightarrow \mathcal{H}(\hat{\boldsymbol{q}}, \mathbf{D}, \theta)$.

$$\mathcal{H}(\hat{\boldsymbol{q}}, \mathbf{D}, \theta) = \mathcal{H}(\hat{\boldsymbol{q}}, \theta) = \left\{ \boldsymbol{q} \in \Delta^L \,\middle|\, \|\mathbf{D}^{-}(\boldsymbol{q} - \hat{\boldsymbol{q}})\| \leq \theta \right\}$$

Similar to Lou et al. (2024), we can construct MVCS problem:

$$\mathbf{D}^*(\boldsymbol{\tau}) = \begin{cases} \inf & \det(\mathbf{D}) \\ \text{s.t.} & \sup\limits_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}} \in \mathcal{F}(\mathbf{X}, \theta)} \mathbb{E}_{\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}} \left[C_j(\boldsymbol{z}_j, d_j) \mid \mathbf{X} \right] \leq \tau_j, \quad \forall j \in [J], \\ & \mathbf{D} \succeq \boldsymbol{r} \end{cases}$$

References

5 Method 2: Mixture DRO Approach

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Problem Statement and Model Formulation

Thank you for listening!
Any questions?